



Manipulating aspen (*Populus tremuloides*) seedling size characteristics to improve initial establishment and growth on competitive sites

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ABSTRACT

Trembling aspen is an important component of the boreal mixedwood forest region and is a key component of numerous ecological functions. As a result, aspen has become a target species for forest restoration after industrial disturbance; however, aspen seedling afforestation research is still in its infancy as seedling and site characteristics and their interactions which drive establishment success in the field are poorly understood. This study examined the growth response of three aspen size classes across six sites differing in competition and soil moisture availability. With competing vegetation present, tall seedlings with lower root-to-shoot ratios had less height growth aboveground and reduced root egress (relative to total root mass) belowground, indicating that light was not the most limiting factor aboveground and that higher root-to-shoot ratios may aid in competition for rooting space. Differences between size classes were less pronounced on sites with no competition or sites with extreme conditions (high competition or low soil moisture availability) during outplanting. Reducing competing vegetation before outplanting improved height growth, particularly in seedlings with high root-to-shoot ratios. For aspen, planting short to medium-sized seedlings with higher root-to-shoot ratios offers a reasonable balance between cost and production of a stress-tolerant seedling.

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Introduction

Recovery and restoration of forest land disturbed by industrial activity requires the successful reestablishment of various ecological functions that govern forest ecosystems (Jacobs et al. 2015). Establishing a tree canopy is an essential step towards this goal, as tree canopies are instrumental towards forest soil development and the associated recovery of a forest understory community (Sorenson et al. 2011; Macdonald et al. 2012). Establishing a forest canopy will also help exclude undesirable weedy herbaceous vegetation that can dominate these systems early on and prevent forest development (Setterfield et al. 2005; Mandryk and Wein 2006). However, other challenging environmental conditions such as low resource availability (water, nutrients, and light), driven by soil conditions and vegetation competition, can lead to a period of prolonged physiological stress for seedlings (Close et al. 2005). This period of reduced growth and survival after outplanting is referred to as “transplant shock”, which is considered the result of seedlings acclimatizing to their new environmental conditions (Sutton and Tinus 1983). Minimizing this period is desirable and can be achieved by manipulating site conditions or seedling planting stock. While transplant shock has generally been associated with restricted root growth or poor root-soil contact preventing the roots from supplying enough water to the shoot (Grossnickle 2005), it can also be attributed to unsuitable seedling characteristics or poor site conditions (Close et al. 2005). The presence of competing vegetation can further contribute to

transplant shock, particularly on older forest reclamation sites, which were not immediately revegetated with tree species after initial disturbance.

The influence of root and shoot competition on planted seedlings is dependent upon site productivity and species composition (Cahill 2002). Grasses generally exert stronger competitive pressure on tree seedlings, relative to forb species (Elliott and White 1987; Gordon et al. 1989; Pywell et al. 2003). The growth form of grasses is characterized by extensive tillering, terminal buds close to the soil surface, and high root-to-shoot ratios (Wilson 1998). This results in strong belowground competition for tree seedlings since seedlings are typically planted at the same soil depth in which most grass roots are concentrated (Cramer et al. 2012). In contrast, forbs typically exert less belowground competition (McCown and Williams 1968; Zangaro et al. 2016) and may be more competitive aboveground by expressing a ruderal growth strategy through rapid germination and early flowering to colonize open spaces and maximize sunlight exposure (Grime 1977).

Trembling aspen seedlings (*Populus tremuloides* Michx.) are particularly sensitive to competition (Powell and Bork 2004a), as grasses and other early successional herbaceous species are quick to colonize open spaces in the boreal forest (Liefers et al. 1993) and can easily outcompete seedlings for belowground resources due to higher root-to-shoot ratios (Tilman 1988; Powell and Bork 2004b). When faced with high competition, seedlings are not only exposed to limited

resources but also potentially to conditions that limit above and belowground growing space, which can reduce overall establishment success (Bockstette et al. 2018). Growth limitations of aspen seedlings have been well documented when grown with competing grasses (Landhäusser and Lieffers 1998; Bockstette et al. 2018). Competing vegetation on reclamation sites can reduce trembling aspen seedling growth for several years after outplanting (van den Driessche et al. 2003; Martens et al. 2007). The accumulation of grass roots and litter near the surface can also have deleterious effects on resource availability, as the sod layer can impede water infiltration, immobilize nutrients, and insulate and keep soils cool (thereby delaying sucker emergence) (Landhäusser and Lieffers 1998; Wilson 1998; Landhäusser et al. 2007).

For aspen planting stock, specific seedling characteristics that are suited to difficult site conditions may be used to mitigate factors limiting growth on some reclamation sites; however, identifying these characteristics is often difficult (Jacobs et al. 2015; Macdonald et al. 2015). Seedling quality is highly dependent on nursery practices that influence morphological and physiological seedlings characteristics which are associated with growth and survival under field conditions (Wilson and Jacobs 2006). For conifer seedlings, height and diameter have been used as indicators of seedling quality and have been shown to improve performance with competing vegetation (Jobidon et al. 1998; South and Mitchell 1999; Jobidon et al. 2003); however, these appear to be poor indicators of first-year growth for aspen seedlings (Martens et al. 2007; Landhäusser et al. 2012a). Martens et al. (2007) observed that naturally regenerated aspen with high root-to-shoot ratios and non-structural carbohydrate (NSC) reserves in roots outgrew nursery produced seedlings that were initially ten times taller, indicating these characteristics may be a more relevant indicator of seedling quality in aspen. Seedlings with high root-to-shoot ratios and root NSC reserves can be developed through the use of shoot growth inhibitors and black-out treatments (Landhäusser et al. 2012a), or stressful conditions such as mild drought or high vapor pressure deficits (Kelly et al. 2015). High root-to-shoot ratios and root NSC reserves have been shown to increase aspen seedling performance on nutrient-poor sites (Landhäusser et al. 2012a, 2012b) and under drought conditions in a greenhouse (Kulbaba 2014); however, it is unclear how height and root-to-shoot ratios influence aspen seedling survival and growth when competing vegetation is present.

In this paper, we evaluated the effect of size and root-to-shoot ratio on the early establishment of aspen seedlings after outplanting over a range of field conditions that varied in the amount and type of competition, as well as in edaphic conditions.

Methods

Seedling production

Aspen seedlings were grown from seeds (seedlot: NNAIT 16-84-23-5-2012 AW) collected in the Peace River Dry Mixed-wood Lowland region (645–655 m a.s.l.) from open-pollinated

sources in 2012. To be conscious of the known variation in seedling production and growing conditions among nurseries, seedlings were grown at two nurseries (Woodmere Nursery Ltd. (WM), Fairview, Alberta (56.064, –118.378; 647 m a.s.l.) and at the University of Alberta (UA), Edmonton, Alberta (53.526, –113.527; elevation 677 m a.s.l.)) using the same procedure. To ensure substrate and soil packing uniformity, the type of substrate (peat) was the same and the filling of containers was done at one facility (Woodmere). Seedling size was manipulated by varying the sowing time and container size. Two sowing times (April and May of 2013) and two container volumes (volume of 340 and 700 ml using a 615A or 815A styroblock containers, respectively; Beaver Plastics Ltd., Alberta, Canada) were used in both nurseries. The styroblock containers were composed of a polystyrene foam block with a series of cells which varied by diameter instead of depth for each specific container size, allowing a similar root structure to be maintained between treatments. All seedlings were fertilized with a custom-made solution of 1 g/L 10-52-10 (N-P-K) with chelated micro-nutrients, that was applied two weeks after seeding then fertilized twice per week with a solution of 1.2 g/L 6-15-10 (N-P-K) plus chelated micronutrients until the soil reached saturation. In June, the concentration of fertilizer was doubled (2.4 g/L) for all seedlings and fertilization was terminated on September 13, 2013. All seedlings were initially grown for a total of 9 weeks in the greenhouse at an average temperature of 25°C, relative humidity of 50%, and 20 h extended daylight photoperiod using artificial lighting. Afterwards seedlings were moved outside for the rest of the growing season (with earlier sow dates moved out first). To maximize root-to-shoot ratios, a shoot growth inhibitor (Paclobutrazol, Bonzi®, Syngenta, North Carolina, USA) was applied to all seedlings (Landhäusser et al. 2012a). The inhibitor was applied by soaking the root systems in a water bath containing a solution of 5 ml of Bonzi per L of water (equivalent to 20 ppm paclobutrazol per liter of water) in mid-July. Bud set occurred approximately two weeks after treatment. After complete leaf senescence occurred (November 1), seedlings were lifted, packaged into cardboard boxes with plastic liners and placed into frozen storage (–4°C) until outplanting in the spring of 2014. At the time of fall lifting, a representative subset of 10 seedlings from several size classes were subsampled to determine initial morphological traits. Height and root collar diameter were measured, and roots and stems were separated for their respective dry

Table 1. Mean pre-planting characteristics of seedling size classes. Means within rows followed by the same letter are not significantly different following analysis of variance on size class groups with initial characteristics (Tukey test, $\alpha = 0.05$). Values in brackets represent one standard error of the mean ($n = 20$). NSC = non-structural carbohydrates.

Size class characteristics	Short	Medium	Tall
Height (cm)	23.8 (1.84) c	46.0 (2.64) b	99.2 (4.84) a
Root collar diameter (mm)	3.47 (0.12) c	4.50 (0.13) b	6.75 (0.29) a
Root mass (g)	2.80 (0.21) c	6.09 (0.47) b	12.8 (1.48) a
Shoot mass (g)	0.64 (0.06) c	1.82 (0.18) b	8.18 (0.97) a
Root-shoot-ratio	4.75 (0.25) a	3.60 (0.27) b	1.60 (0.09) c
Total NSC reserves (%)	46.7 (0.71) a	45.0 (0.73) a	47.3 (1.02) a

mass measurements from which root-to-shoot ratios were calculated (Table 1).

As expected, the variation in growing conditions (particularly during the outdoor period) between the two nurseries resulted in significant variation among seedlings grown with different sowing times and container sizes. Seedling groups (defined by the specific nursery, sowing time, and container size to grow each seedling) were sorted from the shortest to tallest group (based on the average initial height of the subsamples) and then paired into three distinct size classes (short (16.1–28.1 cm); medium (30.3–47.9 cm); tall (72.4–98.6 cm)) for a more robust comparison. Both the short (WM-Apr7-615A and WM-May22-615A) and medium (WM-May1-615A and WM-Apr7-815A) size class were composed of seedling groups produced at WM in 615A or 815A styroblocks sown in April or May. The tall size class consisted of two groups (UA-Apr7-615A and UA-Apr7-815A) developed at UA in both styroblock sizes sown in April. The three size classes also differed in mass allocation to roots relative to the stem (seedlings were dormant and did not have leaves), with the smaller seedling group having the highest root-to-shoot ratio of 4.75 and the tall seedlings with the lowest root-to-shoot ratio of 1.60 (Table 1). It should be noted that a root-to-shoot ratio of 1.60 can be considered high, relative to commercial seedling stock that has been produced in the recent past (root-to-shoot ratio of control seedlings previously produced by Landhäusser et al. (2012a) as low as 0.8).

Site description

In the spring of 2014, seedlings were outplanted into two study areas. The first study area was comprised of three separate and independent sites at the Crop Diversification Center North (hereafter, CDC), a large agricultural research facility in the parkland region of Alberta (Edmonton, Alberta 53.642, –113.361; elevation 651 m a.s.l.) (Table 2). According to previous levels of plant competition, three sites were selected that were relatively uniform in site conditions: (1) an open agricultural field, that had been in crop production for more than 20 years, was kept vegetation-free using plastic mulch and served as a competition-free control (hereafter Mulch); (2) an active large hayfield that had been in production for more than 20 years (hereafter Grass), and (3) an area of the same hayfield that had been plowed just prior to planting in the spring of 2014 (hereafter Plow). In the Mulch site, the planting rows were covered with a 1.8 m wide plastic mulch and periodically weeded in-between rows to minimize competition for the duration of the study.

The second study area (Table 3) was comprised of three sites in the Peace River region in NW Alberta that represented specific site conditions relevant for this study: (1) an agricultural field (hereafter referred to as Ag-field) located 19 km north of Peace River (56.391, –117.209; elevation 518 m a.s.l.), (2) a reclaimed well site (hereafter referred to as Well-site) situated 69 km northwest of Peace River within the boreal mixedwood forest (56.816, –115.707; elevation 572 m a.s.l.), and (3) a reclaimed gravel pit (hereafter referred to as Gravel-pit) located 40 km northeast of Peace River (56.302, –116.561; elevation 683 a.s.l.). Prior to planting, the Ag-field

Table 2. Site characteristics and mean soil characteristics for surface soils collected at Crop Diversification Centre. Different letters within rows indicate significantly different means between soil underneath the mulch, and soil in the Plow and Grass sites following one-way analysis of variance (Tukey test, $\alpha = 0.05$). Values in brackets represent one standard error of the mean (soil texture, $n = 5$; bulk density, $n = 10$; volumetric water content and soil temp, $n = 4$). Soil temperature and volumetric water content reflect measurements taken in 2015 and at a depth of 15 cm.

	Mulch	Plow	Grass
Geographic location	53.646, –113.356	53.648, –113.351	53.648, –113.351
Elevation (m)	651	651	651
Soil adjustment	None	Plowed spring 2014	None
Soil texture (%)	Sandy loam	Sandy clay loam	Clay loam
Sand	67.5 (0.77) a	47.0 (0.38) b	41.2 (0.97) c
Silt	19.5 (1.20) b	26.5 (1.04) a	25.2 (1.69) a
Clay	13.0 (1.69) c	26.5 (1.20) b	33.6 (2.14) a
Bulk density (g cm⁻³)			
0–5 cm	1.25 (0.02) a	0.84 (0.04) b	0.72 (0.04) c
10–15 cm	1.24 (0.04) a	0.96 (0.03) b	0.93 (0.02) b
20–25 cm	1.26 (0.05) a	1.00 (0.04) b	0.96 (0.02) b
Volumetric Water Content (m³_{water} m⁻³_{soil})			
May	0.19 (0.01) c	0.24 (0.01) b	0.28 (0.01) a
June	0.18 (0.01) c	0.21 (0.00) b	0.29 (0.00) a
July	0.19 (0.01) c	0.21 (0.01) b	0.29 (0.01) a
August	0.17 (0.00) c	0.20 (0.00) b	0.26 (0.01) a
Soil Temperature (°C)			
May	12.6 (1.79) a	10.6 (1.38) b	10.4 (1.32) b
June	17.7 (0.98) a	15.7 (0.77) b	15.6 (0.86) b
July	20.6 (0.77) a	18.2 (0.43) b	17.9 (0.38) b
August	18.4 (0.85) a	16.5 (0.77) b	16.7 (0.70) b

was rototilled (spring 2014) and the Well-site was decom-pacted (fall 2013) using a RipPlow (McNabb et al. 2012). No surface site preparation occurred on the Gravel-pit site. A spot application of glyphosate (Roundup, The Scotts Company LLC., Ohio, USA) was used to control Canada thistle (*Cirsium arvense*) in the first summer (2014) in the Ag-field, while no chemical control was used on the other two sites. Seedlings in the Ag-field were within a fenced area, while the Well-site and Gravel-pit were not fenced. Sites in both study areas contained one large planting area representative of the site conditions studied. This area was divided into 10 replicate blocks to account for potential spatial variation within a planting area (though each study site was relatively uniform), with each block containing 8 seedlings (subsample) from each size class spaced 1 m apart within each block and 2 m spacing between blocks. For the purposes of this study, these blocks were treated simply as replicates (rather than random effects) in the subsequent statistical analyses described below.

While the six independent sites between the Edmonton and Peace River study areas do not allow for direct comparisons due to differences in climate and edaphic factors, similarities in site preparation among each site were beneficial for observing establishment success in relation to seedling size and plant competition, as a gradient of plant competition was present between sites in both study areas. With surface site preparation prior to and competition control occurring after outplanting at the Mulch and Ag-field, these sites showed seedling growth in response to reduced competition. At the Plow and Well-site, surface site preparation occurring just prior to outplanting allowed seedlings to respond to competition control in just the first year. In contrast, a lack of site

Table 3. Site characteristics and mean soil characteristics for surface soils collected at sites in the Peace River study area. Different letters within rows indicate significantly different means between sites following one-way analysis of variance (Tukey test, $\alpha = 0.05$). Values in brackets represent one standard error of the mean (soil texture, $n = 5$; bulk density, $n = 10$; volumetric water content and soil temp, $n = 4$). Soil temperature and volumetric water content reflect measurements taken on 2015 at a depth of 15 cm.

	Ag-Field	Well-site	Gravel-pit
Geographic location	56.392, -117.209	56.816; -117.707	56.303; -116.561
Elevation (m)	518	572	683
Soil adjustment	Roto tilled spring 2014	Rip plowed fall 2013	None
Soil texture (%)	Silty Clay Loam	Silty Clay Loam	Loam
Sand	16.0 (0.69) b	18.6 (0.25) b	44.5 (3.61) a
Silt	53.0 (1.13) a	42.8 (0.88) b	39.5 (2.32) b
Clay	31.0 (1.03) b	38.6 (0.64) a	16.0 (1.63) c
Bulk density (g cm⁻³)			
0-5 cm	1.24 (0.05) a	1.19 (0.05) a	1.23 (0.05) a
10-15 cm	1.30 (0.03) b	1.30 (0.03) b	1.66 (0.05) a
20-25 cm	1.37 (0.07) b	1.30 (0.04) b	1.72 (0.06) a
Volumetric Water Content (m³_{water} m⁻³_{soil})			
May	0.40 (0.01) a	0.41 (0.01) a	0.25 (0.02) b
June	0.37 (0.01) a	0.36 (0.01) b	0.18 (0.01) c
July	0.40 (0.01) a	0.32 (0.01) b	0.16 (0.01) c
August	0.36 (0.02) a	0.30 (0.01) b	0.15 (0.01) c
Soil Temperature (°C)			
May	14.4 (2.10) a	11.4 (1.65) b	12.6 (1.18) ab
June	19.6 (1.44) a	16.4 (1.03) b	16.2 (1.01) b
July	21.2 (0.82) a	18.0 (0.65) b	18.0 (0.68) b
August	18.6 (1.00) a	15.6 (0.84) b	16.1 (0.80) b

preparation at the Grass and Gravel-pit sites showed the impact of competing vegetation present during outplanting.

Climate data for the years 2004–2015 at each site has been generated with the *ClimateWNA* v5.50 software package (Wang et al. 2012) available at <http://cfcg.forestry.ubc.ca/projects/climate-data/climatebcwna/#ClimateWNA> and based on a methodology described by Hamann et al. (2013). The period from 2004 to 2013 was used to calculate a longer-term average, whereas the years 2014 and 2015 describe the weather during the experiment. The following nine climate parameters were used to explore seedling responses: MAT (Mean annual temperature), MSP (Mean summer temperature), MCMT (Mean coldest month temperature), MWMT (Mean warmest month temperature), MAP (Mean annual precipitation), MSP (Mean summer precipitation), PAS (Precipitation as snow), and CMD (Hargreaves climatic moisture deficit) (Appendix 1). Hargreaves climatic moisture deficit is the sum of the monthly difference between atmospheric evaporative demand and precipitation (i.e. a measure of water needed in millimeters from sources other than rain (e.g. soil moisture, irrigation) to prevent drought conditions) (Hargreaves 1975).

Site measurements

Daily soil water content and soil temperature, averaged from hourly measurements, were taken with EC-5 (Peace River only) and 5TM sensors (Decagon Devices Inc., Washington, USA) attached to EM50 data loggers (Decagon Devices Inc., Washington, USA) and installed at the center of each planting area in May 2015 (Tables 2 and 3). To assess soil bulk density, two soil cores were collected from the middle point of the 10

blocks at three depths (0–5, 10–15, 20–25 cm) in each planting area. As planting areas at each site were relatively uniform, samples of two blocks were combined for each depth to determine soil texture and nutrient concentrations. Soil texture was determined with particle size analysis using a graduated cylinder and hydrometer (Carter and Gregorich 2008). Soil nutrient samples (Appendix 2) were analyzed by the Natural Resources Analytical Laboratory (Department of Renewable Resources, University of Alberta, Edmonton, Canada) using an ECS 4010 Elemental Combustion System CHNS - O (Costech Analytical Technologies Inc., Valencia, California, USA) to determine NO₃-N and NH₄-N using the 2N KCl method (Benton Jones 2001), and P, K⁺, Mg²⁺ and S concentrations using the microwave-assisted nitric acid digestion method (Hassan et al. 2007).

Above and belowground plant competition was assessed at each site. The aboveground competition was estimated in the first week of July along a transect, placed diagonally across the entire planting area, with a 0.5 × 0.5 m quadrat placed at the intersection point with each block. In each quadrat, plants were identified to the species level, sorted according to their functional group (i.e. grass or forb), and percent cover was visually estimated to the nearest whole percentage. The height of each plant was measured to the nearest centimeter. Each plant was then clipped to determine the dry mass of each species present in the quadrat. The total root mass of competing vegetation was assessed in all planting areas using soil cores collected with an auger at three depths (0–10, 10–20, 20–30 cm). The aspen root component in these soil cores was estimated to be negligible, as cores were collected at distances from seedlings (50 cm in Plow, Grass, Well-site, and Gravel-pit sites and 1 m away in the Mulch and Ag-field sites) which were greater than the average lateral root length measured on the excavated seedlings two weeks prior. To estimate belowground root density of the competing vegetation, roots were separated from soil cores using the procedure described by Teste et al. (2012). Briefly, soil cores were first passed through a 2 mm sieve and coarse root fragments were extracted. The remaining fine root mass was estimated by suspending the soil and organic matter passed through the 2 mm sieve in water. As the fine organic matter was separated from the soil and water using a 250 µm sieve, a subsample was taken, and any fine root fragments present were plucked from the subsample. The mass ratio of fine root to extraneous material in the subsample was then scaled up to the total organic mass to determine total fine root mass. Total root mass was dried at 70°C until weight constancy and weighed to the nearest 0.01 g.

Seedling measurements

Initial height and root collar diameter were measured on all seedlings after outplanting. Initial root-to-shoot ratios for outplanted seedlings were estimated using predictive equations based on the relationship between root collar diameter and root and shoot mass measurements of subsampled seedlings measured prior to planting (Appendix 3). Regression models used to predict root ($R^2 = 0.90$) and shoot ($R^2 = 0.97$) mass were fitted with *Root Collar Diameter* and *Seedling Group* as main effects in addition to the interaction effect

(Appendix 4). As this analysis occurred prior to size classes being defined, measurements from each seedling group (specified by nursery, sowing date, and container size) were used. Significant main and interaction effects produced a unique intercept and slope for each seedling group in the model output (Appendix 3, 4), allowing root and shoot mass to be calculated from the root collar diameter of seedlings within each seedling group.

Total height and height growth after the first and second growing seasons were measured in late August 2014 and 2015. Annual height growth was measured as the shoot extension from the previous bud scar to the terminal bud tip of the main stem. To assess growth allocation above and belowground of each size class after two growing seasons, sixty seedlings were excavated from each site (i.e. two seedlings from each size class per block) in the third week of July 2015. All leaves were removed from seedlings and dried at 70°C for two days to determine total dry leaf mass. Specific leaf area was estimated by measuring the leaf area of ten dried leaves collected from each seedling (LI-3100C, LI-COR, Nebraska, USA) and determining their dry mass. The roots of each planted seedling were carefully excavated using hand trowels, following each root to minimize breakage and root loss. Root systems were separated from the shoots, carefully washed, and dried at 70°C for two days. To assess root system development, roots that had grown outside of the original root plug (egressed roots) were removed from the remaining root plug and dry mass for each were measured separately. Percent egress was measured as the proportion of root mass outside of the seedling plug relative to the total root mass (i.e. egress root mass ÷ total root mass). New root growth was calculated as the difference between total root mass and the initial root mass estimated for each seedling based on the initial root collar measurements (Appendix 3).

Data analysis

Data analysis and visualization has been carried out using the R Language for Statistical Computing (R Core Team 2018) and the *ggplot2* package (Wickham 2016). Representative seedling measurements used in the analysis were averaged from 8 subsampled seedlings from each size class within the 10 blocks at each site (i.e. single measurement averaged from 8 subsampled seedlings per size class per block). For the purposes of this study, these blocks were treated simply as replicates in the subsequent statistical analyses described below. The dataset was analyzed as a 2-way ANOVA using the *gls()* function in the nlme package (Pinheiro et al. 2014) with *Site* and *Size Class* as main effects as well as their interaction. The advantage of using the *glis()* function is that in case of heteroscedasticity in the residuals the model can account for this by allowing each factor level its own variance structure. Estimated marginal means and Tukey-adjusted multiple mean comparisons were carried out using the *emmeans()* function in the *emmeans* package (Lenth 2016). The probability of seedling survival (alive/dead) could only be summarized descriptively since some size class and site combinations had 100% survival leading to the problem of complete

separation. While there are methods available to deal with complete separation (Heinze and Schemper 2002), we did not proceed with statistical significance testing since the results were not sensible. Instead, we have tabulated raw survivorship data. Principal component analyses were carried out using the *PCA()* function in the *FactoMineR* package (Lê et al. 2008) to visualize the multivariate climate space and competition experienced at each test site.

Results

Site conditions

The weather conditions for the duration of the experiment and between the two study areas were mainly driven by differences in precipitation, where the CDC sites in Edmonton had greater summer precipitation and were warmer relative to the sites in the Peace River region (Figure 1(a); Appendix 1, 5). Overall, the weather was colder and wetter during the year of establishment (2014) and warmer and drier in the second year (2015) (Appendix 5). A 10-year climate average (2004–2013) preceding the year of establishment did not reveal any adverse conditions during the year of planting however, large inter-annual fluctuations in weather were present (Appendix 1, 5).

Edaphic conditions varied somewhat among the three CDC sites, where soil texture ranged from a sandy loam to clay loam (Table 2). The Mulch site had higher soil bulk density and soil temperature compared to the other two sites (Grass and Plow); the warmer soil temperature in the Mulch site was likely due to the exposed black plastic mulch warming the ground, whereas the other CDC sites had dense coverage of vegetation insulating the soil (Table 2). Monthly volumetric water content was significantly lower in the Mulch and was highest in the Grass site, with the Plow site falling in-between (Table 2). Some soil macronutrients (NH_4^+ , K, S, Mg) were lower in the Mulch site relative to the Plow and Grass site (Appendix 2).

For the Peace River sites, the Gravel-pit had the highest sand content resulting in a loam soil texture compared to silty clay loam soil textures for the Ag-field and Well-site (Table 3). Soil bulk density was very similar among sites, however the Gravel-pit site had high soil compaction at the lower soil depths ($>1.6 \text{ g cm}^{-3}$) (Table 3). Monthly volumetric water content over the growing season was less than half in the Gravel-pit planting site compared to the Ag-field and the Well-site, however, water content decreased both in the Well-site and Gravel-pit later in the growing season. Seasonal soil temperature was highest at the Ag-field (Table 3). Soil macronutrient concentrations in surface soils varied among the Peace River sites, but there was no systematic trend (i.e. no site that clearly had lower overall soil nutrient concentrations) (Appendix 2).

The type of competition among sites (above and below-ground) was highly variable (Figure 1(b)); however, the principal component analysis also showed a positive correlation between root mass with grass cover and between forb mass and the height of the competing vegetation. The Pearson correlation coefficients for these relationships were both

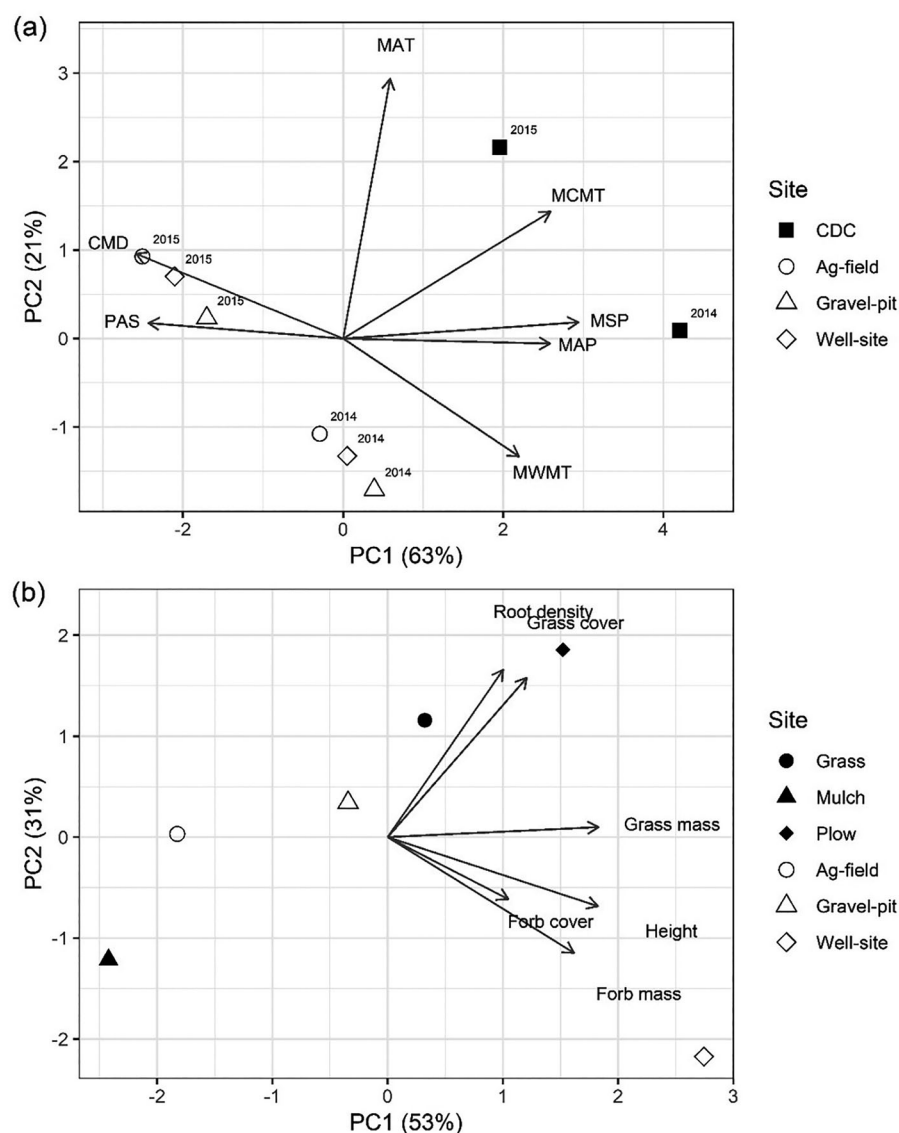


Figure 1. Principal component analyses for (a) weather data in 2014 and 2015 at four site locations and (b) competition parameters at all six sites. CDC = Crop Diversification Centre study area which includes the Grass, Mulch and Plow sites. MAT = Mean annual temperature (°C); MCMT = Mean coldest month temperature (°C); MSP = Mean summer precipitation (mm); MAP = Mean annual precipitation (mm); MWMT = Mean warmest month temperature (°C); PAS = Precipitation as snow (mm); CMD = Hargreaves climate moisture deficit (mm).

significant with $r=0.95$ and $r=0.92$, respectively. On the extremes, the Well-site was dominated by forbs (predominantly *Melilotus officinalis* (L.) Pall.), the Gravel-pit by grass cover (predominantly *Bromus ciliatus* L.) and the Ag-field with the least amount of grass cover. The CDC sites tended to have a mixture of forbs (*Cirsium arvense* (L.) Scop. dominant on the Plow site and *Medicago sativa* L. dominant on the Grass site) and grasses (primarily *Alopecurus aequalis* Sobol., *Bromus inermis* Leyss., *Phalaris arundinacea* L., and *Poa pratensis* L.) with vegetation cover values inside the extremes observed on the Peace River sites (see [Appendix 6](#) and [7](#) for a full species list).

Seedling growth characteristics

Survival at all sites was generally very high (75% to 100%). The highest mortality was observed in the Plow and Gravel-pit sites ([Table 4](#)), specifically for the tall seedlings at the

Table 4. Summary statistics of the seedling survival data after two years ($n = 10$). Min = minimum value; P25 = 25th percentile; P75 = 75th percentile; Max = maximum value.

Site type	Size class	Min	P25	Median	P75	Max
Mulch	Short	1	1	1	1	1
Mulch	Medium	1	1	1	1	1
Mulch	Tall	0.875	1	1	1	1
Plow	Short	1	1	1	1	1
Plow	Medium	0.5	0.75	0.875	1	1
Plow	Tall	0.375	0.75	0.938	1	1
Grass	Short	0.625	0.875	1	1	1
Grass	Medium	0.875	0.875	0.938	1	1
Grass	Tall	0.875	0.906	1	1	1
Ag-field	Short	1	1	1	1	1
Ag-field	Medium	0.75	1	1	1	1
Ag-field	Tall	0.75	0.781	0.875	1	1
Well-site	Short	0.875	1	1	1	1
Well-site	Medium	0.875	0.906	1	1	1
Well-site	Tall	1	1	1	1	1
Gravel-pit	Short	0.625	0.75	0.938	1	1
Gravel-pit	Medium	0.625	0.781	0.875	0.969	1
Gravel-pit	Tall	0.375	0.625	0.75	0.844	0.875

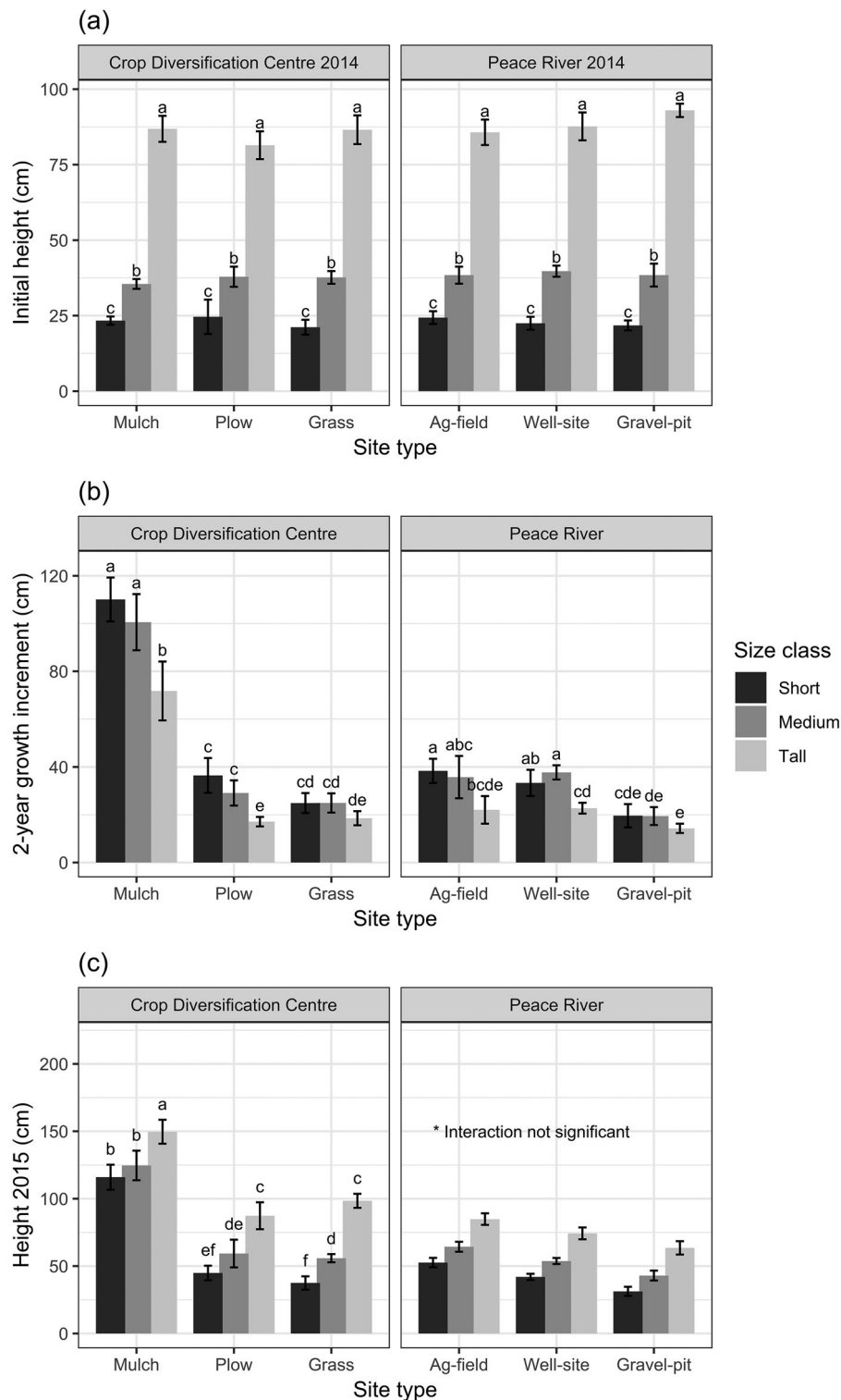


Figure 2. Least squares means of seedling aboveground growth characteristics for size class, site, and study area; (a) initial height in 2014, (b) 2-year growth increment after growing season 2014/15, and (c) total height in 2015. Error bars represent the 95% confidence interval ($n = 10$). Lettering indicates significant differences for significant interaction term within each sub-panel (Tukey-adjusted). In cases where the interaction is not significant refer to Table 5 for significant main effects. Short = Short seedlings; Medium = Medium tall seedlings; Tall = Tall seedlings.

Gravel-pit site (median value = 75%, Table 4). Short seedlings tended to have the highest survival with four of the six sites showing 100% survival with exception of the most extreme sites (Grass site at CDC and Gravel-pit in Peace River, Table 4). However, these differences could not be tested statistically since the logistic regression analysis resulted in

complete separation (see Materials and Methods section for details).

At the time of planting the height of seedlings in the tall size class was over three times that of the medium and short size class (Figure 2(a)). Initial seedling height had a strong influence on subsequent height growth. On four out

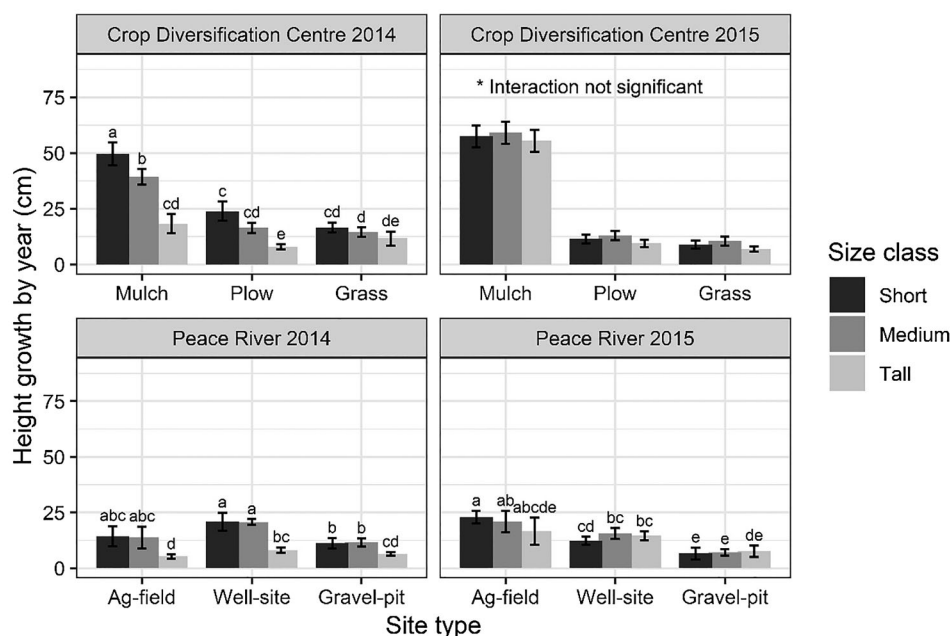


Figure 3. Least squares means of aspen height growth for size class, site, year, and study area. Error bars represent the 95% confidence interval ($n = 10$). Lettering indicates significant differences for significant interaction term within each sub-panel (Tukey-adjusted). In cases where the interaction is not significant refer to Table 5 for significant main effects. Short = Short seedlings; Medium = Medium tall seedlings; Tall = Tall seedlings.

of the six sites (Mulch, Plow, Ag-field, Well-site) short size class seedlings grew more in height (height increment) than the tall size class over the two growing seasons (Figure 2(b)) while the medium size class grew more on half the sites. After two growing seasons, the initial height advantage of the tall size class was reduced to 30–40% (Figure 2(c)). When separating

the growth of seedlings by growing season, short and medium seedlings had greater growth in the first growing season (2014) compared to the tall seedling size class; however, in the second growing season height growth was similar among the size classes (Figure 3). Total leaf mass was highest on sites with less competing vegetation (Mulch and Ag-field) and the tall size class seedlings had greater leaf mass than the shorter seedlings across sites (except on the Gravel-pit) (Figure 4(a)). There was no difference in specific leaf area between sites or size classes at CDC; however, specific leaf area was lowest at the Ag-field (Figure 4(b)).

Overall, the tall size class had significantly more root growth than the short size class in the Mulch and Grass sites (CDC) as well as the Ag-field and Well-site (Peace River study area) (Figure 5(a)). However, proportionally the relative increase in total root mass was often (though dependent on the site) greatest (four time increase from initial conditions) in the shortest size class relative to the medium and tall size class, where root mass was only 2.2–2.5 times greater after two years. Root growth was proportional to total root mass, as tall seedlings showed greater root growth relative to small seedlings at each site (Figure 5(b)). Root-to-shoot ratios after two growing seasons were very similar among size classes with most of the differences occurring among sites, except for the Grass and Gravel-pit sites where the short size class continued to have significantly higher root-to-shoot ratios relative to the tall size class (Figure 5(c)). Seedlings in the Mulch site (least competition) showed the lowest root-to-shoot ratios of all sites (Figure 5(c)). Comparing the three sites at CDC, the percent root egress was highest in the Mulch (least competition). Between the Peace River sites, the level of root egress was extremely low at the Gravel-pit relative to the other sites; it is notable that the short size class showed the greatest

Table 5. Table of significant main effects for the studied traits.

Height 2015 (cm) Peace River					
Size class	Mean	SE	df	lower 95% CL	upper 95% CL
Tall	74.2 a	1.96	27.9	70.2	78.2
Medium	53.7 b	1.16	35.5	51.3	56.0
Short	41.9 c	0.98	30.0	39.9	43.9
Site type	Mean	SE	df	lower 95% CL	upper 95% CL
Ag-field	67.3 a	1.46	27.6	64.3	70.3
Well-site	56.7 b	0.99	39.1	54.7	58.6
Gravel-pit	45.9 c	1.63	27.0	42.6	49.2
Height growth by year (cm) Crop Diversification Centre 2015					
Size class	Mean	SE	df	lower 95% CL	upper 95% CL
Tall	54.0 b	0.92	44.3	22.1	25.8
Medium	27.5 a	1.13	44.9	25.2	29.8
Short	25.9 ab	1.03	36.1	23.8	28.0
Site type	Mean	SE	df	lower 95% CL	upper 95% CL
Mulch	57.3 a	2.30	25.4	52.6	62.0
Plow	11.3 b	0.72	29.4	9.81	12.8
Grass	8.80 c	0.53	31.9	7.72	9.87
Root egress (%) Peace River					
Site type	Mean	SE	df	lower 95% CL	upper 95% CL
Ag-field	25.3 a	1.20	27.7	22.9	27.8
Well-site	16.1 b	1.03	28.0	14.0	18.2
Gravel-pit	5.48 c	0.57	25.5	4.30	6.66
Specific leaf area (cm ² /g) Crop Diversification Centre					
Size class	Mean	SE	df	lower 95% CL	upper 95% CL
Tall	136 a	2.49	33.0	131	142
Medium	127 b	2.30	30.2	122	131
Short	131 ab	2.56	28.7	126	136
Site type	Mean	SE	df	lower 95% CL	upper 95% CL
Mulch	125 b	1.65	26.4	122	129
Plow	133 ab	3.08	26.5	127	139
Grass	136 a	3.17	27.0	130	143

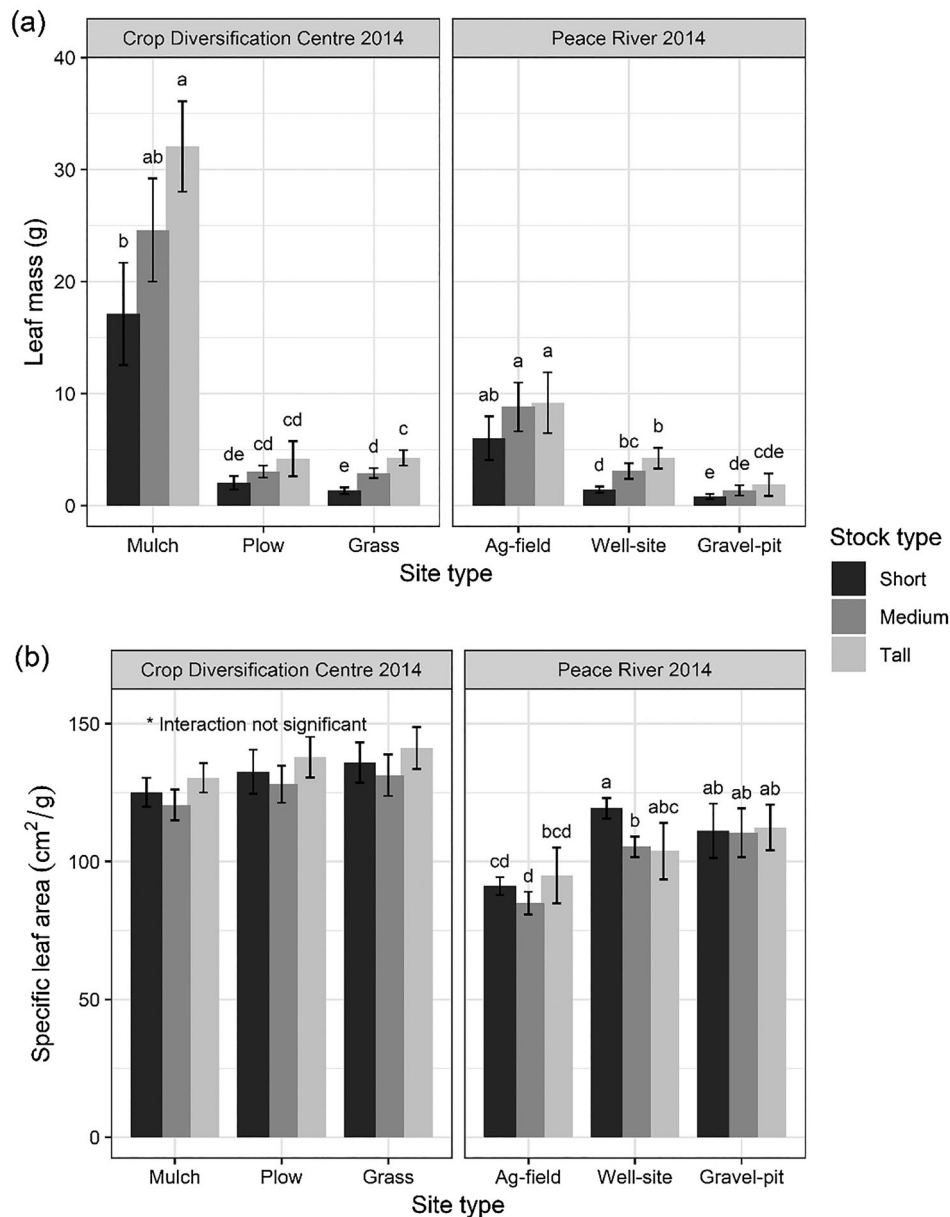


Figure 4. Least squares means of aspen seedling leaf measures for different size classes, sites, and study area in the second growing season; (a) total leaf mass and (b) specific leaf area. All measurements were taken in 2015. Error bars represent the 95% confidence interval ($n = 10$). Lettering indicates significant differences for significant interaction term within each sub-panel (Tukey-adjusted). In cases where the interaction is not significant refer to Table 5 for significant main effects. Short = Short seedlings; Medium = Medium tall seedlings; Tall = Tall seedlings.

root egress (relative to total root mass) on both the Ag-field and Well-site (Figure 5(d)).

Discussion

Regardless of the degree and composition of vegetation competition, the potential advantage of using a taller aspen seedling did not result in greater overall growth benefits (in the first 2 years) relative to using the short and medium-sized seedlings. This response may be at least partly attributable to morphological differences between the tall and short seedling size classes, including higher initial root-to-shoot ratios, that might have allowed for greater root egress into the surrounding soil providing more water for shoot and leaf expansion. High initial root-to-shoot ratios have been previously

shown to result in greater height growth (Martens et al. 2007; Landhäusser et al. 2012a; Landhäusser et al. 2012b), while greater transpirational demand placed on taller seedlings (with more leaf area and lower root-to-shoot ratio) planted into a stressful environment may have further contributed to reduced shoot extension (Grossnickle 2005). Seedling size classes had similar specific leaf area within each site in both study areas; however, seedlings in the tall size class typically had greater total leaf mass (excluding Ag-field and Gravel-pit). This would suggest that total leaf area was higher in taller seedlings, resulting in greater transpirational demand. When faced with aboveground competition, aspen seedlings tend to prioritize shoot growth (Bockstette et al. 2018). In our study, shoot growth appeared to be prioritized in the shorter seedlings potentially to prevent them from

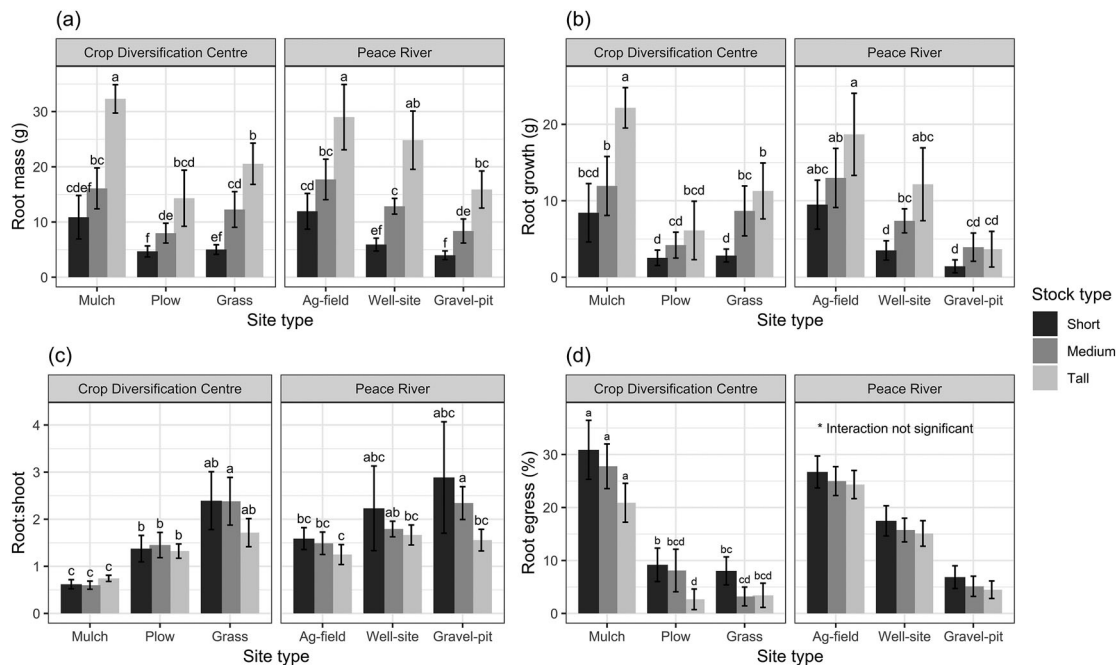


Figure 5. Least squares means of aspen seedling belowground measures for different size classes, sites, and study area after two growing seasons; (a) root mass, (b) root growth, (c) root to shoot ratio, and (d) root egress. All measurements were taken in 2015. Error bars represent the 95% confidence interval ($n = 10$). Lettering indicates significant differences for significant interaction term within each sub-panel (Tukey-adjusted). In cases where the interaction is not significant refer to Table 5 for significant main effects. Short = Short seedlings; Medium = Medium tall seedlings; Tall = Tall seedlings.

being shaded out, while taller seedlings were less directly affected by aboveground competition and as a result prioritized carbon allocation to other parts of the plant. In fact, greater total root growth was found in taller seedlings in all sites except the Gravel-pit (Figure 5(b)). The tall seedlings also had initially larger root systems with a greater NSC reserve pool (Table 1), which could have supported higher root growth (Figure 5(b)). In addition, light availability has been shown to be an important factor for improving root development of aspen seedlings (Landhäusser and Lieffers 2001; Kelly et al. 2015). Increased water stress (from belowground competition) may also be a contributing factor towards increased root growth for improving internal water balance (Ritchie and Dunlap 1980), responses similar to those were found in Aleppo pine seedlings (*Pinus halepensis* Mill.) planted with and without competition (Cuesta et al. 2010).

Competition removal prior to planting appeared to benefit the short and medium-size classes, which showed greater first-year height increment over tall size class seedlings with low root-to-shoot ratios. Although these plants were smaller, their higher root-to-shoot ratios (Table 1) were likely beneficial in the development of vigorous shoot growth. In aspen, high root-to-shoot ratios are often a good indicator of carbon storage reserves (Landhäusser et al. 2012a). At each of these sites (Plow, Well-site), vegetation competition was removed temporarily following site preparation allowing shorter seedlings with a high root-to-shoot ratio to grow faster. However, this window of opportunity was short-lived, as in the following year, competing vegetation recovered rapidly with different grasses and tall weedy forbs such as sweet clover (*Mellilotus officinalis* (L.) Pall.) and Canada thistle (*Cirsium arvense* (L.) Scop.) proliferating. The rapid

development of competing vegetation depressed height increment of short and medium-sized seedlings in the second year, leading to similar second year height growth for all three size classes (Figure 3). Despite this, total growth increment over the two-year period was still significantly greater in the short size class relative to tall size class on both the Plow and Well-site due to differences in root-to-shoot ratio, further illustrating the influence of initial morphology of seedling stock on initial establishment (Grossnickle 2012). Achieving early canopy closure is a management strategy for controlling competing vegetation in forest restoration (Macdonald et al. 2015) and is driven by factors such as species composition, planting density, and early growth. Regardless of these drivers, early seedling growth is crucial and may require the management or control of vegetation beyond the first year, especially on very competitive sites.

While tall size class seedlings maintained their height advantage at every site, a tall shoot was not critical to the growth of aspen on sites with favorable growing conditions (i.e. the Mulch site). The difference in total height (i.e. height advantage) of the tall size class was drastically reduced (from 300% to approximately 30–40%) after two growing seasons. Similarly, the shortest size class showed the greatest increase in total root mass (four-time increase from initial conditions), nearly doubling the increase in root mass observed in the medium and tall size class (2.2–2.5 times greater after two years). Reduced growth response differences driven by morphological characteristics had previously been observed in aspen on less stressful sites (Landhäusser et al. 2012b). The lack of competition was likely the main driver of aspen growth response on the Mulch site, as the soil conditions appeared to be generally more favorable in the Plow and

Grass sites (i.e. lower bulk density and higher soil nutrient concentrations, see [Appendix 2](#)). On the Ag-field site; however, where competition was also low and the moisture deficit was high ([Figure 1\(a\)](#)), tall seedlings did not grow substantially in height (less than 25 cm over 2 years) compared to the two shorter size classes. In this situation, tall size class seedlings had the greatest quantity of leaf area to maintain ([Figure 4](#)) with the lowest relative root mass, potentially placing it at greatest risk of drought and further emphasizing that moisture stress can have as much of an impact on seedling development as competing vegetation. This result also highlights the potential drawback of using taller aspen seedlings where moisture may be an environmental constraint.

On sites where significant competition was present from the onset of outplanting (Grass and Gravel-pit sites), size played less of a role in the growth and establishment of aspen. The abundance of root competition in the Grass site or the very dry conditions at the Gravel-pit diminished the benefit of a high root-to-shoot ratio and all seedlings performed poorly. The presence of competing roots has been associated with limiting resource uptake in newly planted seedlings (Bockstette et al. 2017), but has also been associated with other effects such as allelopathy (Landhäusser and Lieffers 1998). Principally, a seedling stock type capable of enhanced root growth should be beneficial on sites with significant belowground competition. However, tall size class seedlings which had the greatest total root growth, did not appear to benefit from this response in our study over the short-term. This may have been attributable to the type of root system development, as the tall seedlings grew relatively fewer roots from the plug into the surrounding soil ([Figure 5\(d\)](#)). Root egress was an important growth parameter in our study as shorter size class seedlings with increased egress showed greater first year height growth. Increased root egress should also lead to better soil plant contact making these seedlings likely better candidates in a competitive environment. The reduced root egress observed in the taller size class could suggest that despite the increased root growth in larger seedlings, most root development did not occur outside of the existing root plug. Similar results have been observed in lodgepole pine seedlings (*Pinus contorta* Douglas), where seedlings with a larger root mass showed reduced egress due to root binding (Hellum 1981). Coupled with a lack of root egress, a taller shoot carrying more leaf mass ([Figure 4\(a\)](#)) was detrimental for tall seedlings, which showed higher mortality at both sites and particularly at the Gravel-pit site during the dry conditions of 2015. While initial seedling morphology can be tailored for drier site conditions to a certain extent (Pinto et al. 2011; Landhäusser et al. 2012b), there are physiological limitations within a given species where manipulation of seedling morphology is not sufficient (Rose et al. 1993) and other species or selected drought tolerant genotypes may have to be used on these sites.

The methods of this study limit its ability to establish causal relationships between seedling size and outplanting performance in relation to site conditions, due to a lack of replication at the site level. However, we successfully observed a general pattern in the effectiveness of a high root-to-shoot ratio for

aspen seedlings, as short size class seedlings had greater two-year height growth over the tall size class on four of the six sites. Additionally, observing similar growth responses between size classes at the Grass and Gravel-pit sites due to competition being present during outplanting highlights the importance of competition control, as extreme growing conditions will diminish any benefit of modified seedling characteristics. By observing similar seedling responses between two study areas which exhibited a gradient of vegetation competition among the different sites, our study was effective in showing seedling establishment in response to seedling size over a wider range of site characteristics rather than narrowing in on one type of treatment.

Conclusions

Contrary to seedling characteristic studies for other species, which recommend taller seedling stock when competing vegetation is present (South and Mitchell 1999; Franklin and Buckley 2009; Pinto et al. 2011), there was not enough evidence from our study to recommend this approach for aspen. While the simultaneous influence of initial height and root-to-shoot ratio could not be separated in this study, taller seedlings were not competitive as they were limited by a lower root-to-shoot ratio. Tall seedlings with a high root-to-shoot ratio may be more competitive with competing vegetation; however, this must be balanced against overall production costs. Although we did not find substantive differences between short and medium seedlings, planting a short seedling (<46 cm height) in a competitive environment could pose risks in a longer time frame (3–5 years) if the seedling is not able to overtop competing vegetation (our study did not address this timeline). Our study did illustrate that short-term competition control was impactful and if this treatment had been maintained for an additional growing season it may have been a sufficient time period to allow seedlings from the short and medium-size classes to better tolerate or even suppress competing vegetation at this planting density. Improving site conditions plays a critical role for seedling establishment, as controlling competing vegetation will help to ensure successful seedling establishment. When competing vegetation is present, aspen seedlings will grow at a much slower rate, regardless of their initial size. Although climate factors cannot be controlled, differences in seedling development among the sites in our study show the importance of moisture availability on these boreal sites; therefore, creating site conditions that enhance root development and moisture availability may better facilitate establishment in dry years.

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Appendices

Appendix 1

Weather data for the year 2014 and 2015 for the sites in the Edmonton and the Peace River study areas. Values in parentheses represent a 10-year average leading up to the experiment (2004–2013). MAT = Mean annual temperature (°C); MWMT = Mean warmest month temperature (°C); MCMT = Mean coldest month temperature (°C); MAP = Mean annual precipitation (mm); MSP = Mean summer precipitation (mm); PAS = Precipitation as snow (mm); CMD = Hargreaves climate moisture deficit (mm).

Year	Study Area	Site	Latitude	Longitude	Elevation	MAT	MWMT	MCMT	MAP	MSP	PAS	CMD
2014	Edmonton	*CDC	53.642	−113.361	651	2.4 (2.7)	18.3 (17)	−9.7 (−14)	486 (493)	369 (363)	95 (92)	164 (199)
2014	Peace River area	Ag-field	56.392	−117.209	518	0.9 (1.3)	18 (16.6)	−13.6 (−17.2)	407 (407)	257 (261)	101 (122)	298 (296)
2014	Peace River area	Gravel-pit	56.303	−116.561	683	0.2 (0.6)	17.4 (15.9)	−14.3 (−17.9)	452 (445)	292 (278)	126 (142)	227 (238)
2014	Peace River area	Well-site	56.816	−117.707	572	0.7 (1.2)	17.9 (16.5)	−13.6 (−17.3)	417 (429)	262 (272)	104 (133)	264 (254)
2015	Edmonton	*CDC	53.642	−113.361	651	4 (2.7)	17.2 (17)	−10 (−14)	447 (493)	318 (363)	87 (92)	276 (199)
2015	Peace River area	Ag-field	56.392	−117.209	518	2.5 (1.3)	16.7 (16.6)	−14 (−17.2)	390 (407)	225 (261)	145 (122)	359 (296)
2015	Peace River area	Gravel-pit	56.303	−116.561	683	1.7 (0.6)	16 (15.9)	−14.6 (−17.9)	436 (445)	254 (278)	160 (142)	280 (238)
2015	Peace River area	Well-site	56.816	−117.707	572	2.3 (1.2)	16.5 (16.5)	−13.9 (−17.3)	407 (429)	231 (272)	156 (133)	313 (254)

* CDC = Crop Diversification Centre study area which includes the Grass, Mulch and Plow sites.

Appendix 2

Mean nutrient concentration from surface soils (0–5 cm) collected from sites at the Crop Diversification Centre and Peace River study areas in 2015. Different letters within rows indicate differences among sites following one-way analysis of variance (Tukey test, $\alpha = 0.05$). Sites at the Crop Diversification Centre were not compared against sites in the Peace River region. Values in brackets represent one standard error of the mean ($n = 5$).

Soil nutrients (mg L ^{−1})	Mulch	Plow	Grass	Ag-Field	Well-site	Gravel-pit
NH ₄ ⁺	1.79 (0.08) b	2.80 (0.15) ab	4.21 (0.95) a	1.30 (0.17)	1.32 (0.24)	2.01 (0.15)
NO ₃	0.48 (0.03)	0.51 (0.05)	0.48 (0.02)	0.38 (0.10) a	0.20 (0.02) ab	0.12 (0.01) b
P	1.92 (0.16)	2.04 (0.08)	2.18 (0.16)	1.22 (0.10) b	1.98 (0.05) a	2.06 (0.12) a
K	4.46 (0.42) b	6.62 (0.47) a	7.02 (0.19) a	8.99 (1.15) a	7.41 (0.22) ab	5.46 (0.32) b
S	1.39 (0.13) b	2.36 (0.21) b	4.69 (0.93) a	1.05 (0.13) b	4.72 (1.19) a	1.32 (0.07) b
Mg	5.75 (0.47) b	8.11 (0.39) b	12.4 (1.09) a	14.3 (0.71) a	15.3 (0.45) a	7.78 (0.55) b

Appendix 3

Slope estimates for predictive equations used to estimate root and shoot mass.

Root mass estimates for each seedling type were based on the following equation:

$$M_{\text{root}} = (2.26 + A_n) + (0.94 + B_n) * RCD$$

where M_{root} is the root mass; A_n is the change to intercept for seedling group n ; B_n is the change to interaction term for seedling group n ; RCD is the root collar diameter.

The following coefficients for each seedling group are as follows:

WM-Apr7-815A: $A_n = 0$ $B_n = 0$

WM-Apr7-615A: $A_n = -2.14$ $B_n = -0.07$

WM-May1-615A: $A_n = -6.01$ $B_n = 1.05$

WM-May22-615A: $A_n = -5.93$ $B_n = 0.89$

UA-Apr7-815A: $A_n = -9.64$ $B_n = 2.32$

UA-Apr7-615A: $A_n = -17.92$ $B_n = 3.07$

Shoot mass estimates for each stock type were based on the following equation:

$$M_{\text{shoot}} = (-3.23 + A_n) + (1.07 + B_n) * RCD$$

where M_{shoot} is the shoot mass; A_n is the change to intercept for seedling group n ; B_n is the change to interaction term for seedling group n ; RCD is the root collar diameter.

The following coefficients for each seedling group are as follows:

WM-Apr7-815A: $A_n = 0$ $B_n = 0$

WM-Apr7-615A: $A_n = 3.07$ $B_n = -0.82$

WM-May1-615A: $A_n = 0.75$ $B_n = -0.14$

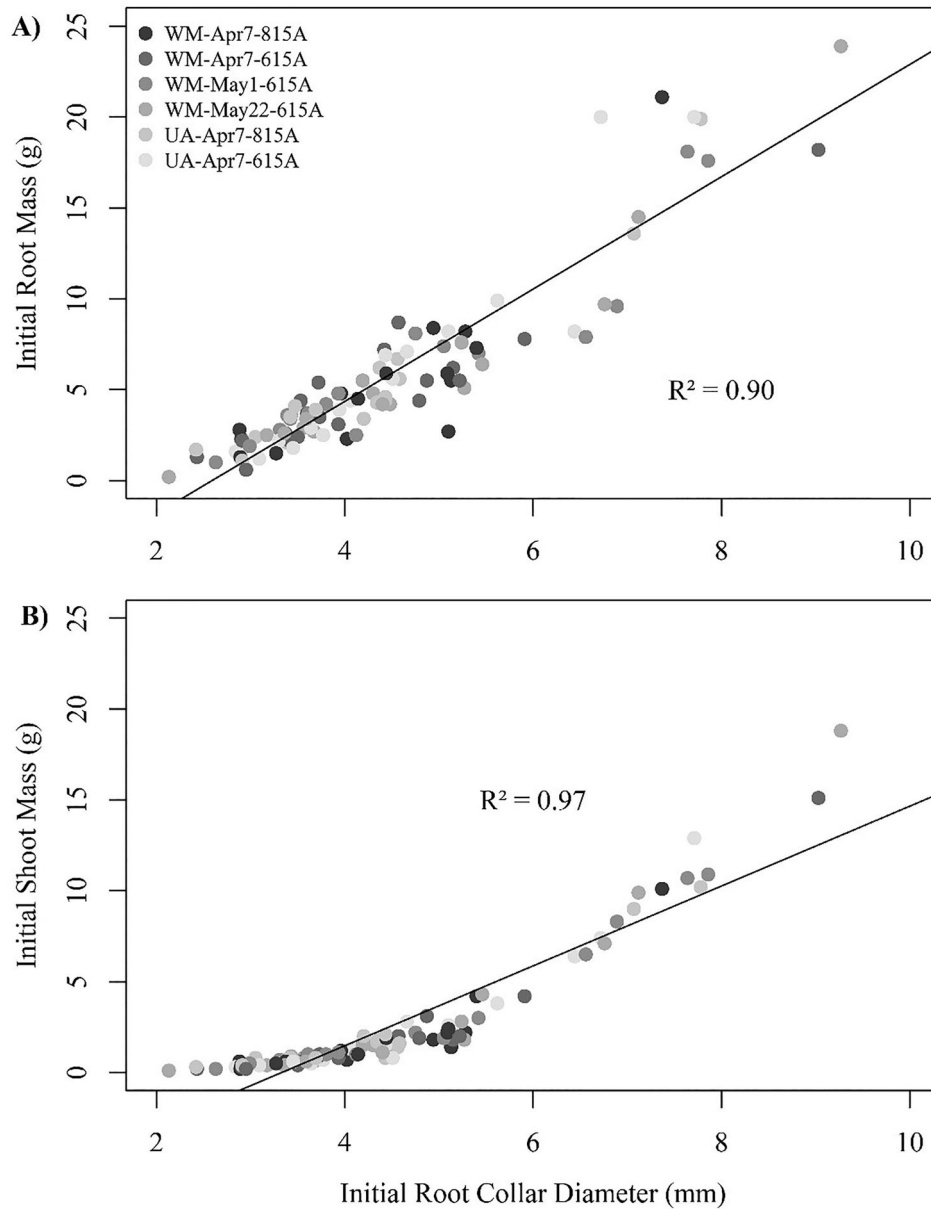
WM-May22-615A: $A_n = 2.44$ $B_n = -0.67$

UA-Apr7-815A: $A_n = -12.24$ $B_n = 2.42$

UA-Apr7-615A: $A_n = -6.28$ $B_n = 1.44$

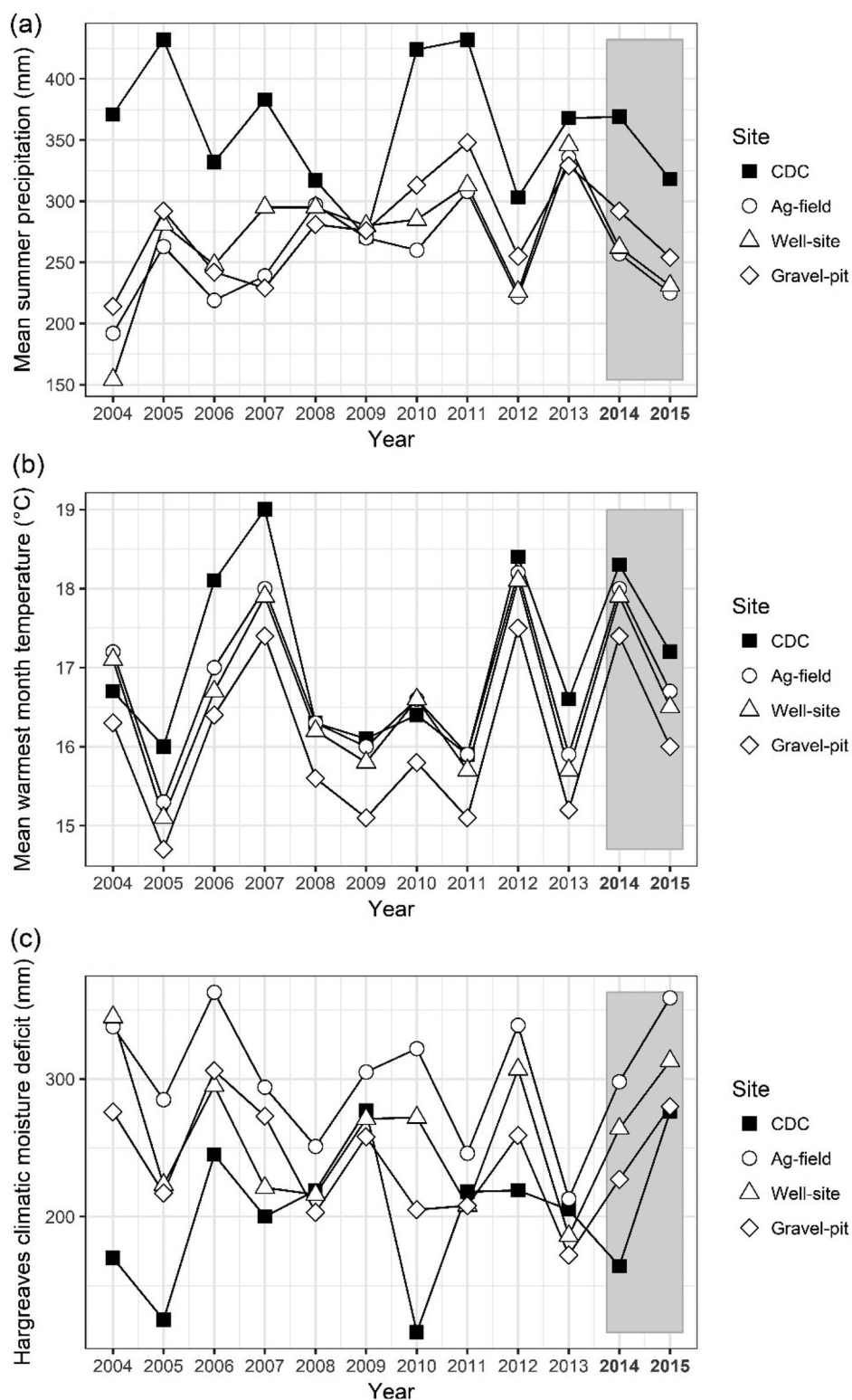
*Root and shoot mass estimates that resulted in negative values or a value that was more than 3 standard deviations away from the mean were omitted.

Appendix 4



Relationship between initial root collar diameter and initial (a) root ($R^2 = 0.90$) and (b) shoot mass ($R^2 = 0.97$) of subsampled seedlings prior to outplanting. Dots color coded according to seedling groups shown in legend. Regression line equations for each seedling group were used to estimate root and shoot mass for outplanted seedlings. Single regression line for WM-Apr7-815A seedling group shown for clarity.

Appendix 5



Historical weather data at each site characterizing the climate leading up to the experiment as well as during the experiment in 2014 and 2015 (gray rectangle).

Appendix 6

Mean percent cover, height, and dry mass of plant species found in the three sites (Mulch, Plow and Grass) at the Crop Diversification Center in Edmonton, AB. Values in brackets represent one standard deviation of the mean ($n = 10$). Dominant species are bolded.

Site	Species	Percent cover	Height (cm)	Mass (g)
Mulch	<i>Agropyron</i> sp.	6.00 (5.21)	11.4 (4.74)	6.56 (5.70)
	<i>Capsella bursa-pastoris</i> (L.) Medik.	1.00 (0.00)	22.3 (21.3)	0.72 (1.00)
	<i>Chenopodium album</i> L.	1.25 (0.50)	15.6 (8.24)	0.65 (0.95)
	<i>Cirsium arvense</i> (L.) Scop.	4.00 (2.65)	12.5 (6.02)	2.88 (2.25)
	<i>Erodium cicutarium</i> (L.) L'Hér. ex Aiton	5.67 (8.08)	8.72 (4.63)	1.89 (3.02)
	<i>Polygonum convolvulus</i> (L.) Å.Löve	2.60 (2.19)	17.7 (7.32)	0.90 (0.84)
	<i>Potentilla norvegica</i> L.	4.50 (4.95)	16.5 (3.54)	2.84 (3.08)
	<i>Senecio</i> sp.	1.67 (0.58)	2.58 (1.23)	0.58 (0.38)
	<i>Taraxacum laevigatum</i> DC.	1.00 (0.00)	7.00 (0.00)	0.10 (0.00)
	<i>Taraxacum officinale</i> F.H. Wigg	7.00 (7.00)	19.8 (4.37)	3.66 (1.52)
	<i>Thlaspi arvense</i> L.	3.22 (3.03)	18.2 (8.93)	3.49 (3.24)
	<i>Cirsium arvense</i> (L.) Scop.	12.4 (8.37)	30.7 (8.66)	10.7 (8.09)
	<i>Fragaria vesca</i> L.	1.00 (0.00)	5.00 (0.00)	0.11 (0.00)
	<i>Medicago sativa</i> L.	5.40 (3.21)	29.2 (6.27)	3.73 (2.47)
Plow	Poaceae sp.*	18.1 (10.3)	60.0 (9.61)	22.6 (11.4)
	<i>Taraxacum laevigatum</i> DC.	1.00 (0.00)	10.5 (0.00)	0.27 (0.00)
	<i>Taraxacum officinale</i> F.H. Wigg	2.11 (1.45)	15.3 (6.32)	0.56 (0.55)
	<i>Trifolium hybridum</i> L.	1.00 (0.00)	10.0 (0.00)	0.04 (0.00)
	<i>Cirsium arvense</i> (L.) Scop.	3.00 (1.73)	9.33 (6.66)	1.54 (1.52)
	<i>Medicago sativa</i> L.	6.14 (3.76)	35.7 (11.4)	7.25 (5.85)
	Poaceae sp.*	15.2 (7.39)	54.3 (14.4)	18.8 (6.86)
Grass	<i>Potentilla anserina</i> (L.) Rydb.	1.50 (0.71)	13.6 (4.60)	0.53 (0.21)
	<i>Taraxacum laevigatum</i> DC.	1.75 (0.50)	13.9 (6.20)	0.44 (0.45)
	<i>Taraxacum officinale</i> F.H. Wigg	4.29 (2.63)	15.6 (5.76)	2.01 (1.62)

*Grasses present in the Plow and Grass sites include: *Alopecurus aequalis* Sobol., *Bromus inermis* Leyss., *Phalaris arundinacea* L., and *Poa pratensis* L.

Appendix 7

Mean percent cover, height, and dry mass of plant species found in Peace River sites. Values in brackets represent one standard deviation of the mean ($n = 10$). Dominant species are bolded.

Site	Species	Percent cover	Height (cm)	Mass (g)
Ag-field	<i>Chenopodium album</i> L.	3.00 (1.41)	28.5 (7.78)	1.97 (1.28)
	<i>Cirsium arvense</i> (L.) Scop.	1.50 (0.71)	4.00 (2.83)	0.24 (0.21)
	<i>Dracocephalum parviflorum</i> Nutt.	10.0 (0.00)	31.0 (0.00)	5.22 (0.00)
	<i>Elymus glaucus</i> Buckley	8.00 (4.24)	58.5 (29.0)	9.31 (3.45)
	<i>Festuca ovina</i> L.	6.00 (0.00)	7.00 (0.00)	5.72 (0.00)
	<i>Festuca saximontana</i> Rydb.	8.40 (7.09)	11.8 (7.98)	7.80 (8.07)
	<i>Lappula echinata</i> Gilib.	8.66 (10.0)	35.3 (22.0)	5.15 (6.98)
	<i>Senecio vulgaris</i> L.	5.00 (0.00)	27.0 (0.00)	2.37 (0.00)
	<i>Sonchus oleraceus</i> L.	1.00 (0.00)	33.0 (0.00)	0.21 (0.00)
	<i>Taraxacum officinale</i> F.H. Wigg	8.14 (6.01)	5.85 (3.63)	1.47 (1.32)
	<i>Thlaspi arvense</i> L.	2.33 (1.53)	13.0 (12.5)	0.91 (1.25)
	<i>Agropyron trachycaulum</i> (Link) Gould ex Shinners	4.80 (2.97)	92.5 (16.5)	16.6 (9.72)
	<i>Bromus inermis</i> Leyss.	3.71 (3.09)	87.7 (7.80)	10.1 (8.44)
	<i>Equisetum arvense</i> L.	15.9 (11.1)	16.0 (3.87)	4.14 (4.00)
Well-site	<i>Melilotus alba</i> Medik.	2.66 (0.58)	73.3 (13.5)	5.07 (3.25)
	<i>Melilotus officinalis</i> (L.) Pall.	16.8 (15.5)	95.8 (61.1)	26.3 (23.3)
	<i>Phleum pratense</i> L.	2.00 (0.00)	72.5 (0.71)	3.14 (1.75)
	<i>Plantago major</i> L.	14.5 (3.54)	36.0 (0.00)	4.13 (0.52)
	<i>Poa pratensis</i> L.	1.60 (0.89)	51.0 (13.6)	1.79 (2.23)
	<i>Rubus idaeus</i> L.	3.00 (0.00)	6.00 (0.00)	0.05 (0.00)
	<i>Sonchus arvensis</i> L.	6.66 (2.89)	16.3 (11.5)	1.79 (1.84)
	<i>Taraxacum officinale</i> F.H. Wigg	2.00 (0.00)	10.0 (0.00)	0.10 (0.00)
	<i>Trifolium hybridum</i> L.	14.6 (8.02)	27.3 (17.0)	7.30 (5.75)
	<i>Trifolium repens</i> L.	2.00 (0.00)	7.00 (0.00)	0.06 (0.00)
	<i>Achillea millefolium</i> L.	4.66 (6.35)	6.33 (2.31)	0.87 (1.33)
	<i>Agrostis scabra</i> Willd.	2.33 (0.58)	32.7 (8.74)	0.80 (0.09)
	<i>Bromus ciliatus</i> L.	9.90 (7.87)	29.7 (9.29)	5.12 (5.97)
	<i>Carex aenea</i> Fern.	4.00 (1.41)	4.00 (1.41)	3.70 (1.51)
Gravel-pit	<i>Cirsium arvense</i> (L.) Scop.	1.33 (0.58)	5.66 (1.53)	0.19 (0.10)
	<i>Equisetum arvense</i> L.	3.50 (1.97)	5.00 (1.41)	0.30 (0.14)
	<i>Equisetum</i> sp.	1.00 (0.00)	3.00 (0.00)	0.10 (0.00)
	<i>Fragaria virginiana</i> Duchesne	14.3 (9.49)	5.80 (1.87)	2.67 (1.45)
	<i>Melilotus officinalis</i> (L.) Pall.	14.9 (13.3)	10.0 (10.1)	3.81 (4.60)
	<i>Plantago major</i> L.	2.00 (0.00)	6.25 (3.18)	0.17 (0.11)
	<i>Rubus idaeus</i> L.	10.0 (0.00)	7.00 (0.00)	1.92 (0.00)
	<i>Salix</i> sp.	1.50 (0.71)	10.5 (7.78)	0.37 (0.10)
	<i>Sonchus arvensis</i> L.	6.00 (0.00)	10.0 (0.00)	0.80 (0.00)
	<i>Taraxacum officinale</i> F.H. Wigg	6.11 (3.02)	8.00 (3.35)	1.31 (1.38)